

Performance of RAG-Based Systems in Industrial Organizations: A Case Study in the Automotive Industry

Nils Mayat

BMW AG

Munich, Germany

ORCID 0009-0007-3327-2490

Charlotte Wachter

Laboratory for Machine Tools and

Production Engineering WZL

RWTH Aachen University

Aachen, Germany

ORCID 0009-0004-7382-4649

Sonja Spatzenegger

BMW AG

Munich, Germany

Email sonja.spatzenegger@online.de

Marcos Padrón Hinrichs

Laboratory for Machine Tools and

Production Engineering WZL

RWTH Aachen University

Aachen, Germany

ORCID 0000-0001-6358-892X

Tim Weißer

BMW AG

Munich, Germany

ORCID 0009-0007-1370-3679

Robert H. Schmitt

Laboratory for Machine Tools and

Production Engineering WZL

RWTH Aachen University

Fraunhofer Institute for

Production Technology (IPT)

Aachen, Germany

ORCID 0000-0002-0011-5962

Abstract— Knowledge is a critical factor for the success and productivity of organizations, especially in the knowledge-intensive automotive industry. Retrieval-augmented generation (RAG) represents a potential approach for knowledge management in organizations, as it combines the flexibility of generation models with the precision and interpretability of knowledge retrieval. But to what extent does knowledge management in the automotive industry benefit from RAG? In a semi-structured interview, the usefulness of RAG was evaluated regarding efficiency and comprehension. An experiment was conducted where participants from direct and indirect sector of the organisation completed four search tasks across three levels of understanding. Both tasks in the participants' area of expertise and non-specialist tasks were used to evaluate the support provided by RAG in different fields. Pairs of participants were used to compare a traditional search approach as a benchmark with the RAG-based search. The findings show that the participants using the RAG-based search applications complete the tasks in significant less time, using queries with higher complexity and searching fewer pages than those using the traditional search approach. Trends regarding the usage of fewer queries were observed. Regarding the comprehension factor, participants using RAG predominantly achieve a higher level of correctness in the search tasks. Across the three levels of understanding, the RAG-based search tools show more constant correctness than the traditional tools. Especially for the most complex tasks, participants using the RAG-based search applications obtain considerably higher correctness. Overall, searching with RAG is perceived as a more positive and satisfying experience. In summary, these results indicate that RAG technology has the potential to benefit knowledge management in the knowledge intensive automotive industry by increasing productivity and helping employees develop a more comprehensive and deeper understanding.

Keywords— RAG, Knowledge Management, Information Retrieval, Automotive

I. INTRODUCTION

The automotive industry is characterized by a knowledge-intensive environment in all phases of the product life cycle, requiring efficient knowledge management processes and technologies [1]. Systems based on large language models

(LLM) with retrieval augmented generation (RAG) have been proposed to overcome the accuracy limitations of LLMs by integrating domain-specific knowledge for knowledge management and information retrieval [2,3]. While some of the proposed approaches to enterprise information retrieval have been investigated [4,5], less research has been carried out in industries that require a high degree of specialized knowledge from different collaborative domains, as in the case of the automotive industry [5]. In particular, the benefits of using RAG-based systems to support information retrieval in highly aggregated knowledge organizations such as automotive OEMs have not yet been sufficiently explored. Therefore, this case study aims to answer the following research question: "How do users with RAG-based systems perform in regard to efficiency, comprehension and perception compared to traditional search tools within a knowledge-intensive industrial organization exemplified by the automotive industry?" The paper investigates this question using an experimental design consisting of semi-structured interviews in an automotive OEM. The paper is organized as follows: Section 2 provides an overview of relevant terms in the field of information retrieval systems, LLMs and RAG. Section 3 describes the methodology for the experiment and the results are presented in Section 4. Section 5 concludes with a discussion and implications of the study results as well as an outlook for further work.

II. STATE OF THE ART

According to Chowdhury an information retrieval system enables the identification and retrieval of relevant information from an organized collection of documents that fully or partially match the user's query [6]. Information retrieval systems form a crucial role in the industrial organizations due to the increasing complexity of production systems and the need to incorporate various, heterogenous data into decision making processes [7]. With the rapid developments in digitalization and the emergence of powerful AI-applications, such as LLMs, knowledge management strategies and information retrieval systems have developed accordingly [8].

A. Large language models for information retrieval

Approaches in recent literature aim to leverage the generative abilities and large general knowledge base of LLMs for information retrieval tasks. Nevertheless, there are limitations to the application of existing LLMs. The issue of model *hallucination*, lack of domain-specific information as well as the static nature of the knowledge base pose hurdles to the integration in knowledge intensive settings, such as the automotive domain [9,10]. Several domain-specific LLMs have been proposed to integrate specific knowledge and reduce the risk of hallucination for the domains of healthcare [11], finance [12] and law [13] through the process of model fine-tuning. While these domain-specific models may increase the performance of knowledge-intensive tasks, the considerable effort which goes into training-dataset curation as well as the high computational costs of the model training pose limitations to its practical application [9,14]. Furthermore, the static nature of the knowledge base is not addressed through fine-tuning [10,14]. An established alternative that aims to address this issue is In-Context-Learning (ICL) in which the required information is stored externally and introduced into the model with the user query, enabling the introduction of up-to date data [3,10]. In this context, Lewis *et al.* developed the RAG architecture, which is comprised of three main parts, indexing, retrieval and generation [2]. *Indexing* involves the partitioning of documents into chunks of a defined length as well as the creation of dense vector representations of the text-chunks through an embedding model. The chunk-vectors are then stored in an external vector store. Once the query is introduced, the *retrieval* part consists of the identification of the closest match of the query embedding to the information in the vector store. The top k vector-chunks are retrieved applying measure of similarity methods such as cosine similarity search. In the *generation* step, both the query and the retrieved chunks are passed into the LLM to generate a response [14,15].

B. Application of RAG for information retrieval in industry

Bahr *et al.* developed a question-answering system based on knowledge graphs and RAG to retrieve information on Failure Mode and Effects Analysis (FMEA) documentation [16]. The ontology-based representation of FMEA information enabled the contextualization of numeric values and allowed for a correct interpretation of risk priority numbers. The RAG-approach demonstrated improvements in the evaluation categories *correctness*, *usability*, *relevance*, *completeness* and *retrieval time* in comparison to a traditional Excel-based information search [16]. Yet, the proposed RAG-approach is limited to the definition of the use-case specific ontology. Álvaro & Barreda implemented a RAG-based question-and-answer application for the quality control in ceramic tile manufacturing [17]. As a knowledge base, defect documentation as well as relevant academic articles were selected. The answers provided by the RAG application exceeded the ones generated by GPT 4.0 in terms of accuracy. The combination of both internal documentation as well as broader industry knowledge allows for the generation of comprehensive and accurate descriptions of potential failure causes and solutions [17]. Yet, evaluations metrics such as precision, comprehension or efficiency were not considered in the publication. A RAG application for the automotive sector was introduced by Hernandez-Salinas *et al.* [18]. The Intelligent Driving Assistance System, IDAS, is a question-and-answer application based on the contents of the vehicles

manuals. The RAG architecture enables the identification and retrieval of relevant manual passages based on the user query as well as the generation of a human-like response. The answers provided by the application were evaluated considering multiple language models of varying sizes regarding answer *correctness*, *relevancy*, *semantic similarity*, *context precision*, *context recall* and *faithfulness*. Smaller models such as gemma-2b, gemma-7b or llama2-7b generally performed poorer than larger models, especially regarding the answer correctness [18]. While the RAG approach provides faithful answers based on vehicle manual information, the user-perspective on the application was insufficiently considered. The RAG approach was not evaluated against conventional manual search in terms of user experience, retrieval effort or retrieval time. Due to the novelty of the RAG architecture, only few use-case specific applications have been researched. The previously mentioned publications attest to the increased accuracy and trustworthiness of RAG-based applications compared to traditional or even LLM-based information retrieval tools. However, existing evaluations lack a comprehensive consideration of the user and application perspective. The benefit of RAG applications for knowledge-intensive industries, specifically in the context of an automotive OEM, has not been sufficiently investigated.

III. METHODS

A. RAG System Designs

The first system is designed to enable user interactions through a combination of search and chat functionalities. The frontend includes the user interface for search and chat interactions as well as the admin interface for system management and data source monitoring. The backend includes several services to perform following tasks, which are necessary for the RAG approach: processing of queries and retrieving information by the *search service*, generating and summarising answers by the *LLM proxy service*, data preparation by the *ingestion service*, *connectors* for integrating external data sources, managing indexed data and system configurations by the *data store* and the *tenant management service* for configuration, multi-tenant support and adding contextual relevance. The second system called “Maintenance Assistant” has access to maintenance topics in production environments. The *knowledge base* consists of maintenance documents from a network drive. The *user interface* allows users to select the search precision level. The

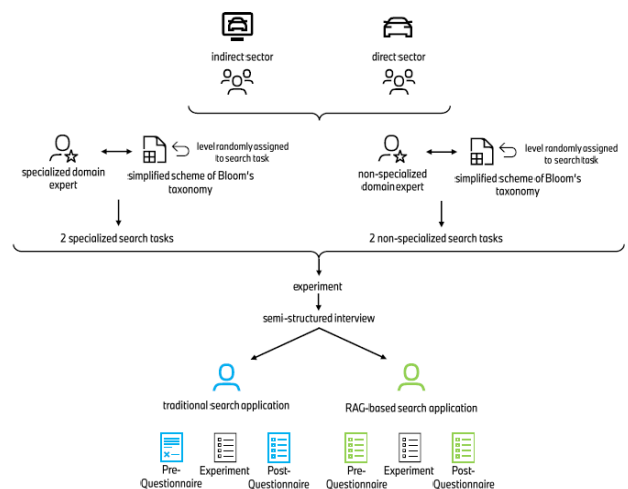


Fig. 1. Experiment design

chat engine condenses the conversation and the user query into a comprehensive question, retrieves relevant context from the knowledge base, and generates a response using the LLM. This process ensures that responses are informed by previous interactions and the broader knowledge base. The traditional search is carried out in the data source underlying the RAG application used for comparability. The underlying data source of the search and chat application is *Confluence*, while the maintenance assistant's data source is the network drive.

B. Experiment Design

The experimental design is shown in Fig. 1. It incorporated a semi-structured interview comprising three main components: a pre-questionnaire, an experiment answering questions using the different tools, and a post-questionnaire.

The pre-questionnaire measured factors such as personal attitudes towards AI and prior experience with AI tools using a five-point Likert scale to capture potential biases [19,20]. Central to the design was the experiment, which involved four single-choice search tasks. This approach was based on prior research to measure efficiency and productivity [19,21]. Participants were paired from different departments of an automotive organization. One participant in each pair used the RAG-based search application, while the other used the respective traditional search method. The experiment document was identical for both participants in each pair and consisted of three segments. The first segment included two search tasks in a single-choice format within the participants specialisation (Sp.), with four answer options per task. The second segment assessed familiarity with topics outside their expertise using a Likert scale. The third segment mirrored the first but focused on tasks outside the participant's expertise (N. Sp.). Participants were asked to find information using the specified tool and make informed decisions to solve the tasks. The experiment was conducted using a MS Word document,

with identical documents provided to both participants in each pair. To account for language differences, the documents were available in either English or German. The estimated completion time for the interview was 60 minutes. Participants unfamiliar with the RAG-based application received a prior briefing on its features.

In order to investigate different type of tasks Bloom's taxonomy was used. Bloom's taxonomy classifies educational objectives and consists of a hierarchical structure that ranges from basic memorization to advanced analytical skills [22]. Adequate to the total size of the experiment, a condensed version of the taxonomy was applied, grouping the six levels into three broader categories: Knowledge & Comprehension (K&C), Application & Analysis (A&A), and Synthesis & Evaluation (S&E). Each task was randomly assigned to one of three levels of understanding. The tasks were developed by an expert from the relevant department to ensure the realism of the activities, the appropriateness of the content and the similarity of the complexity of the tasks [20]. Participants' comprehension and efficiency were assessed by evaluating the correctness rates and analysing the search process. In the post questionnaire the participants were asked to provide subjective ratings on their perceived experience with the tool. The questions were either on a scale or dichotomous format. An interval scale in the form of a five-point Likert scale was used. For the traditional search tools, the post-questionnaire included 11 questions addressing the dependent variables: cognitive load, self-efficacy, usability, performance, trust, quality, and understanding. It concluded with an optional free text field. The post-questionnaire for the RAG-based search tools contained the same 11 questions as the version for traditional search tools, with a minor modification to the last question of the quality variable. The statement "I was satisfied with the quality of the search results." was changed in the

TABLE I. TASK RESULTS

Metric	Task	Median		Mean(SD)		Wilcoxon Signed-Rank Test		
		RAG	Trad.	RAG	Trad.	W	p-value/ adjusted p-value	r _{rb}
Search Time (s)	Overall	189.5	270.5	248.07(185.36)	382.83(283.17)	851.50	0.0015/ 0.0093*	0.2910
	Sp.	210.5	244.0	291.34(217.46)	344.42(216.15)	301.00	0.3135/ 1	0.4062
	N. Sp.	175.0	292.5	204.79(136.05)	421.24(335.84)	142.50	0.0009/ 0.0057*	0.1923
	K&C	147.0	238.0	153.81(81.58)	365.62(278.16)	56.00	0.0018/ 0.0107*	0.1624
	A&A	216.0	308.0	308.65(238.25)	359.88(217.52)	136.00	0.3157/ 1	0.3875
	S&E	225.0	297.0	284.54(165.89)	426.33(350.77)	104.00	0.1974/ 1	0.3467
Number of Queries	Overall	2	2	2.99(2.83)	3.80(4.17)	848.50	0.3672/ 1	0.2900
	Sp.	2	2	3.21(3.38)	3.13(3.34)	240.50	0.8827/ 1	0.3246
	N. Sp.	2	2.5	2.76(2.17)	4.47(4.81)	179.50	0.1772/ 1	0.2422
	K&C	2	3	2.46(1.50)	4.77(4.73)	74.50	0.0904/ 0.5423	0.2123
	A&A	1	2	3.54(4.02)	2.85(2.87)	106.00	0.7388/ 1	0.3020
	S&E	2	2	2.96(2.33)	3.79(4.63)	92.00	0.9036/ 1	0.3067
Mean query complexity	Overall	29.75	6.50	43.50(40.57)	7.28(4.72)	2926.00	<0.0000/ <0.0000***	1
	Sp.	31.75	6.44	38.17(23.82)	6.76(4.90)	741.00	<0.0000/ <0.0000***	1
	N. Sp.	28.42	6.88	48.84(52.05)	7.79(4.55)	741.00	<0.0000/ <0.0000***	1
	K&C	23.50	6.71	25.41(12.64)	7.32(4.26)	351.00	<0.0000/ <0.0000***	1
	A&A	41.12	7.37	42.14(29.37)	6.68(4.32)	300.00	<0.0000/ <0.0000***	1
	S&E	31.75	6.18	62.58(57.02)	7.78(5.57)	351.00	<0.0000/ <0.0000***	1
Number of pages visited ^a	Overall	0	4	1.55(3.05)	10.22(19.03)	247.00	<0.0000/ <0.0000***	0.0844
	Sp.	1	4	2.13(3.43)	6.89(8.76)	102.00	0.0003/ 0.0016**	0.1377
	N. Sp.	0	4.5	0.97(2.54)	13.55(25.19)	35.00	<0.0000/ <0.0000***	0.0472
	K&C	0	4.5	0.73(1.15)	11.58(20.86)	2.50	<0.0000/ 0.0002***	0.0071
	A&A	0	3.5	1.96(3.83)	8.72(12.52)	33.00	0.0005/ 0.0029*	0.0940
	S&E	1	5	2.00(3.44)	10.88(23.04)	59.50	0.0166/ 0.0993	0.1983

^a. A minimum value of 0 was reached, when the participant answered the question only with the answer of the RAG and without looking at proposed further information of the RAG

RAG questionnaire to “I was satisfied with the quality of the generated answers.” to better align with the generative nature of the RAG-based search tools. Additionally, this version included a second question on understanding, evaluating the understandability of the generated text, and two questions on tool preference, assessing participants' willingness to use the tool in their everyday work in the future and to recommend it to others. Furthermore, participants were asked how they perceived the unique features of the RAG applications to determine whether these features affected the experimental outcomes. The questionnaire ended with an optional free text field for additional feedback, similar to the version for traditional search tools.

C. Subjects

The semi-structured interviews were conducted either in person at the participants' workplace or remotely via Microsoft Teams. The experiments were recorded using the screencast tool *OBS Studio* (64bit). Adhering to the automotive organization's guidelines on data security and participant selection, no demographic data was collected. For each data point, three individuals were involved: two test subjects and one expert for search task development. The study included a total of 38 test subjects and nine experts. Participants were drawn from both the direct and indirect areas of the automotive company. The direct area includes functions directly contributing to the creation of the company's primary products or services, such as manufacturing and production, while the indirect sector encompasses supporting functions not directly involved in production, such as IT and administrative roles. Moreover, the sample represented employees from two different areas within the automotive company: production (24 participants) and development (14 participants).

D. Data Analysis

The participants' search processes was analysed using screencasts to document the search. Metrics include search duration, number of pages visited, queries submitted, and query complexity (QC), calculated as:

$$QC = 0.5 * (n_{token}) + 0.5 * (l_{token}), \quad (1)$$

with n_{tokens} being the number of tokens and l_{token} being the token lengths [21]. Each metric was analysed overall tasks, over Sp. And N.Sp. tasks and over each of the three levels of understanding. Before each test a Shapiro-Wilk test was performed to check for normal distribution. Since at least one of the distributions for each test (meaning the distribution of the results using RAG and using the traditional tool) showed a non-normal distribution the Wilcoxon signed rank test was selected. The p value was adjusted for multiple testing using the Bonferroni correction. Additionally, the rank-biserial correlation (r_{rb}) was reported as an effect size. The correctness of participants' answers was documented. The correctness was tested using the McNemar's test and again the p values were adjusted using Bonferroni. Additionally, the odds ratio (OR) was reported [23]. The pre- and post-questionnaire's answers were transformed into numerical values for Likert scales (1 to 5) and dichotomous scales (0 for “No” and 1 for “Yes”). Attitudes towards AI, experience with AI tools, and familiarity with search topics were analysed to check for biases. Spearman correlation ρ as calculated to assess the impact of these factors on search times and answer correctness

[23]. Statistical analysis was carried out using the python packages *scipy* [24] and *statsmodels* [25].

IV. RESULTS

A. Efficacy

The results are presented in Tables I, categorized by various examination criteria, including overall performance, specialization, and level of understanding. Each task's outcome is represented across three rows for each metric; for instance, the search time for a task is included in the overall analysis, the specialization analysis, and the analysis of the level of understanding (e.g. K&C). Statistical significance is denoted by asterisks, indicating p-values below 0.05, 0.01, and 0.001, respectively. P-values of one were obtained due to the Bonferroni correction, which involved multiplying the original p-value by six, corresponding to the six tests conducted per metric. The adjusted p-values were capped at 1. A significant difference was observed in search times across all tasks ($W = 851$, $p < 0.01$, $r_{rb} = 0.2910$). Participants utilizing the RAG system required a mean search time of 248.07 seconds ($SD = 185.36$ seconds) to 382.83 seconds ($SD = 283.17$ seconds). Furthermore, significant differences were noted in search times for the N. Sp. tasks ($W = 142.50$, $p < 0.01$, $r_{rb} = 0.1923$) and the K&C tasks ($W = 56.00$, $p < 0.05$, $r_{rb} = 0.1624$). In contrast, for other task categories, including A&A, S&E, and specialized tasks, both the mean and median search times were lower when using the RAG tool. Regarding the number of queries issued, no significant differences were detected. Although the mean number of queries was lower for RAG users (2.99, $SD = 2.83$) compared to traditional tool users (3.80, $SD = 4.17$), this trend was not consistent across all task categories. Notably, for specialized tasks, the mean number of queries was slightly higher for RAG users (3.21, $SD = 3.38$) compared to traditional users (3.13, $SD = 3.34$), as well as for tasks categorized under the Analysis & Application level of understanding (3.54, $SD = 4.02$ versus 2.85, $SD = 2.87$). As anticipated, the mean query complexity was significantly greater for RAG users. Since every query issued to the RAG system had a higher complexity than the corresponding queries made using traditional search tools, r_{rb} is one. This can be explained by users issuing search statements using full sentences and not only keywords. The number of pages visited was significantly lower when employing the RAG system. Across all tasks, the mean number of pages visited was 1.55 ($SD = 3.05$) for RAG users compared to 10.22 ($SD = 19.03$) for traditional search tool users, underscoring the efficacy of RAG systems in retrieving relevant information. However, for tasks categorized under the S&E level of understanding, no significant difference was observed, despite substantial differences in the mean and median number of pages visited. Notably, there were instances where the RAG tool's response sufficiently addressed the query, rendering additional searches unnecessary, as evidenced by a median of zero pages visited.

TABLE II. PERCENTAGE OF CORRECTNESS

Task	RAG [%]	Trad. [%]	McNemar's Test		
			$\chi^2(I)$	p-value/ Adjusted p-value	OR
Overall	82.89	68.42	5	0.0266/ 0.1596	3.20
Sp	89.47	73.68	3	0.1460/ 0.8760	3.00
N. Sp.	76.32	63.16	2	0.1797/ 1.0000	3.50
K&C	88.46	73.08	1	0.2188/ 1.0000	5.00
A&A	88.46	92.31	2	1.0000/ 1.0000	0.67
S&E	70.83	37.50	1	0.0215/ 0.1289	9.00

B. Comprehension

The results are systematically presented in Table II. No statistically significant differences in correctness were observed between the two subject groups. However, a trend indicating superior performance by RAG users was noted. Across all tasks, RAG users demonstrated a higher accuracy rate compared to subjects utilizing traditional tools, achieving 82.89% correctness versus 68.42%. Notably, in the A&A tasks, subjects using traditional tools outperformed RAG users, with accuracy rates of 88.46% compared to 82.31%. Conversely, in tasks categorized under S&E, RAG users surpassed traditional users by more than 30%. The calculated OR further support this trend, predominantly yielding values greater than 1, indicating a higher likelihood of correctness among RAG users relative to their traditional counterparts.

C. Additional Factors

The prequestionnaire investigated whether personal attitude towards AI or prior experience with AI tools impacted efficiency and comprehension using RAG-based search technology. The results showed very weak and statistically insignificant correlations between users' attitudes towards AI and search times (Spearman's $\rho = 0.096$, $p = 0.695$), and between prior experience with AI tools and search times ($\rho = 0.025$, $p = 0.920$). The correctness of answers in non-specialized tasks showed no significant correlation with prior topic familiarity, for both RAG-based ($\rho = 0.007$, $p = 0.969$) and traditional search tools ($\rho = 0.193$, $p = 0.246$). Therefore, the improved performance of the RAG-based search tools in terms of efficiency and correctness is likely due to the capabilities of the tools themselves, rather than other external factors.

D. Perception

In the post-questionnaire, participants rated various aspects of their search experience with both traditional and RAG-based search tools. The results indicated that the perceived cognitive load was significantly lower for the RAG-based tools (average rating of 1.95) compared to traditional tools (average rating of 2.79). This suggests that the RAG-based tools are perceived as less cognitively demanding. Participants also reported higher self-efficacy when using the RAG-based tools, with an average rating of 4.05, compared to 3.16 for traditional tools. The usability of the RAG-based tools was rated higher, suggesting they are easier to use and enhance efficiency. The mean rating for tool performance was slightly higher for the RAG-based tools (3.84) compared to traditional tools (3.21). Trust in the results was similar for both tools, with an average rating of 3.53. The perceived reliability was slightly higher for traditional tools (4.0) compared to RAG-based tools (3.74). Participants rated the RAG-based tools as more useful for both easy (4.42, 4.05) and complex tasks (3.95, 3.16). The perceived understanding was significantly better with the RAG-based tools, with an average rating of 4.37 compared to 3.16 for traditional tools, indicating that the RAG-based tools enabled users to gain a deeper understanding. Overall, participants expressed a strong positive intention to use the RAG-based tools in the future, with an average rating of 4.58, and all participants recommended the tool. The free feedback highlighted the need for a greater variety of databases and usability improvements for the RAG-based tools. Traditional tools received more

critical feedback, particularly regarding inconsistent data quality, suboptimal search algorithms, and poor usability for non-specialized tasks. In summary, the RAG-based search tools were favoured for their lower cognitive load, higher self-efficacy, better usability, and improved understanding.

V. DISCUSSION

The research provides a comprehensive evaluation of RAG-based search tools compared to traditional search methods in the automotive industry. The findings highlight several critical advantages of RAG-based tools, underscoring their potential to transform search experiences and enhance productivity in professional settings.

The efficiency analysis reveals that participants using RAG-based applications completed their search tasks 41.18 % faster than those using traditional search methods. The median total search time for the traditional approach was 27.97 minutes, compared to 16.45 minutes for the RAG-based applications, translating into a median time saving of 11.52 minutes. This considerable reduction in search time can be attributed to the ability of RAG-based tools to deliver more direct access to relevant information, minimizing the need for users to sift through multiple pages.

Additionally, the RAG-based approach resulted in fewer queries being issued, particularly in tasks outside the participants' area of expertise. This suggests that these tools are especially beneficial when users lack domain-specific knowledge, reducing the cognitive load associated with search tasks. Although no statistically significant performance improvement was detected, a trend towards a higher rate of correct answers was noted among users of the RAG-based tool. Only 17.12 % of search tasks were answered incorrectly using RAG-based applications, compared to 31.58 % with traditional search methods. The slight decrease in correctness for non-specialized tasks may be due to participants over-relying on the RAG tools and the increased cognitive load associated with higher query complexity. Nonetheless, the overall higher correctness rate emphasizes the effectiveness of RAG-based tools in supporting users' comprehension and reducing errors.

The research also delves into the depth of understanding facilitated by the two search approaches. The RAG-based search condition maintained high correctness levels even at the most complex S&E level. This indicates that the advanced natural language understanding, and knowledge-based reasoning capabilities of RAG-based tools are particularly beneficial in supporting higher-order cognitive activities investigated in this experiment. In contrast, traditional search tools proved less effective at higher levels of understanding, especially in non-specialized domains. This disparity suggests that domain expertise plays a crucial role in the effectiveness of keyword-based search methods, highlighting the limitations of traditional approaches in supporting complex information needs. User perception further supports the advantages of RAG-based search tools. Participants reported lower cognitive load and higher self-efficacy when using RAG-based applications, indicating a more supportive and empowering user experience. Despite these advantages, users did not perceive RAG-based tools as definitively superior in terms of performance, trust, and perceived quality of search outputs. However, the RAG tools received significantly higher ratings for perceived support in achieving deeper

understanding, consistent with the higher comprehension results observed. The positive user feedback and strong intention to use and recommend RAG-based tools suggest their significant potential in improving organizational knowledge management.

Despite the promising results, the research acknowledges several limitations. Data security restrictions prevented the analysis of demographic factors such as gender and age, potentially limiting the generalizability of the findings. The chosen method may not fully capture the complexity of human comprehension. Additionally, the uneven distribution of participants from the direct (26 %) and indirect (74 %) sectors does not necessarily represent the actual user distribution.

VI. CONCLUSION

The case study aimed to evaluate the performance of users with RAG-based search approaches relative to traditional search methods in a knowledge-intensive industry, specifically in the context of an automotive OEM.

The results show that users with the RAG-based search tool outperformed the traditional approach, particularly for non-specialized tasks, exhibiting faster search times, fewer queries, and fewer pages visited. However, for tasks that fall in the class of application and analysis, participants seemed more comfortable with the traditional search method, potentially due to familiarity. The results indicate that practitioners developing RAG-based systems in the industry should target these systems especially for users working with knowledge outside of their specialisation. While the findings of this case study are promising, the researchers acknowledge that the results do not allow for a generalised statement on the superiority of the performance of RAG based systems in an industrial context.

Future research could explore the impact of factors like gender and age on the adoption and effectiveness of RAG-based tools. Furthermore, more extensive methods could be considered to analyse the human computer interaction. Compared to a single agent system as analysed in this study, the rise of multi agent systems might increase the capabilities of RAG systems and therefore the usefulness for knowledge management in industry.

VII. REFERENCES

- [1] F. Ferreira, J. Faria, A. Azevedo & A. L. Marques, "Product lifecycle management in knowledge intensive collaborative environments: An application to automotive industry" *International Journal of Information Management*, 37(1), 1474-1487, 2017
- [2] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.", 2020
- [3] W. Fan, Y. Ding, L. Ning, S. Wang, H. Li, D. Yin et al., "A Survey on RAG Meeting LLMs: Towards Retrieval-Augmented Large Language Models", *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining New York, NY, USA: ACM*, 2024, pp. 6491-6501
- [4] G. Şahin, K. Varol and B. K. Pak, (2024). "LLM and RAG-Based Question Answering Assistant for Enterprise Knowledge Management" In 2024 9th International Conference on Computer Science and Engineering (UBMK) (pp. 1-6). IEEE.
- [5] M. Arslan, S. Munawar and C. Cruz (2024). "Business insights using RAG-LLMs: a review and case study". *Journal of Decision Systems*, 1-30.
- [6] G.G. Chowdhury, "Introduction to modern information retrieval", 3rd ed., London, Facet Publ., 2010. Clerk Maxwell, *A Treatise on Electricity and Magnetism*, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68-73.
- [7] L. Kim, E. Yahia, F. Segonds, P. Véron, and A. Mallet, "i-Dataquest: A heterogeneous information retrieval tool using data graph for the manufacturing industry", *Computers in Industry*, vol. 132(103527), 2021
- [8] M.S. Kolyasnikov, N.R. Kelchevskaya, "Knowledge management strategies in companies: Trends and the impact of Industry 4.0." *Upravlenets – The Manager*, vol. 11, no. 4, pp. 82-96, 2020
- [9] S. Wu, Y. Xiong, Y. Cui, H. Wu, C. Chen, Y. Yuan et al., "Retrieval-Augmented Generation for Natural Language Processing: A Survey.", July 2024, Available: <http://arxiv.org/pdf/2407.13193v2>.
- [10] K. L. Xia, C. Li, C. Zhang, S. Liu, and P. Zheng, "Leveraging error-assisted fine-tuning large language models for manufacturing excellence." *Robotics and Computer-Integrated Manufacturing*, vol. 88(102728), 2024.
- [11] K. Singhal, S. Azizi, T. Tu, S.S. Mahdavi, J. Wei, H.W. Chung, N. Scales, A. Tanwani, H. Cole-Lewis, S. Pfohl et al., "Large language models encode clinical knowledge." *Nature*, vol. 620(7972), pp. 172-180, 2023
- [12] S. Wu, O. Irsoy, S. Lu, V. Dabravolski, M. Dredze, S. Gehrmann, P. Kambadur, D. Rosenberg, and G. Mann, "Bloomberggpt: A large language model for finance.", 2023, Available: <https://arxiv.org/abs/2303.17564>
- [13] Q. Huang, M. Tao, Z. An, C. Zhang, C. Jiang, Z. Chen, Z. Wu, and Y. Feng, "Lawyer llama technical report.", 2023, Available: <https://arxiv.org/abs/2305.15062>
- [14] O. Ovadia, M. Brief, M. Mishaeli, and O. Elisha, "Fine-Tuning or Retrieval? Comparing Knowledge Injection in LLMs.", 2024
- [15] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi et al., "Retrieval-Augmented Generation for Large Language Models: A Survey.", 2024, Available: <http://arxiv.org/pdf/2312.10997v5>
- [16] L. Bahr, C. Wehner, J. Wewerka, J. Bittencourt, U. Schmid, and R. Daub, "Knowledge Graph Enhanced Retrieval-Augmented Generation for Failure Mode and Effects Analysis", 2024, Available: <https://doi.org/10.48550/arXiv.2406.18114>
- [17] J.A. Heredia Álvaro and J.G. Barreda, "An advanced retrieval-augmented generation system for manufacturing quality control.", *Advanced Engineering Informatics*, vol. 64(103007), 2025.
- [18] L.-B. Hernandez-Salinas, J. Terven, E.A. Chavez-Urbiola, D.-M. Córdova-Esparza, J.-A. Romero-González, A. Arguëlles et al., "IDAS: Intelligent Driving Assistance System Using RAG.", *IEEE Open Journal of Vehicular Technology*, vol. 5, pp. 1139-1165, 2024
- [19] S. Noy and W. Zhang. Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, 381(6654):187-192, 2023
- [20] J. Liu. Deconstructing search tasks in interactive information retrieval: A systematic review of task dimensions and predictors. *Information Processing & Management*, 58(3):102522, 2021
- [21] S. E. Spatharioti, D. M. Rothschild, D. G. Goldstein, and J. M. Hofman. Comparing traditional and llm-based search for consumer choice: A randomized experiment. *arXiv preprint arXiv:2307.03744*, 2023.
- [22] B. S. Bloom, M. D. Engelhart, E. J. Furst, W. H. Hill, and D. R. Krathwohl. *Taxonomy of educational objectives: The classification of educational goals. Handbook I: Cognitive domain*. New York: David McKay, 19
- [23] M. Lovric "International Encyclopedia of Statistical Science", Berlin, 2011, doi: 10.1007/978-3-642-04898-2
- [24] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, et al., "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python," *Nature Methods*, vol. 17, no. 3, pp. 261-272, 2020, doi: 10.1038/s41592-019-0686-2.
- [25] S. Seabold and J. Perktold, "statsmodels: Econometric and statistical modeling with python.", *Proceedings of the 9th Python in Science Conference*. 2010.